Five Sensors, One Model

Multimodal activity recognition under a 100 MB constraint — CUHK-X Challenge 2026, Small Model Track

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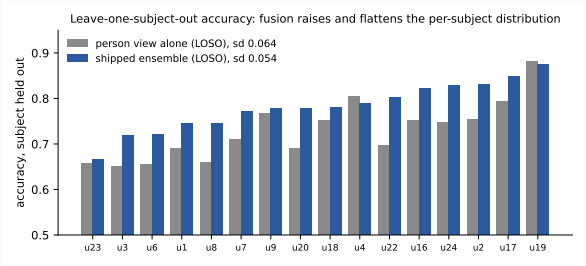
Selection Stage finals, 2026

The problem, in one sentence

Forty everyday activities, from sensors that cannot identify a face, on people the model has never seen, in a single file of at most 100 MB.

- infrared + depth video, thermal video, tracked skeletons, five wearable inertial units, radar timing
- 18 training subjects, 4 test subjects, further held-out subjects evaluated by the organisers
- metric: top-1 accuracy; every inference-time weight inside the 100 MB

Why this is hard: subjects, not clips, are the unit



- one video model, one subject held out at a time: accuracy ranges over **28 points**
- four public subjects: a public score carries ± 6.9 clips at one standard error
- the residual error is *per-subject and correlated*: twin recordings share the identical wrong label 68 % of the time

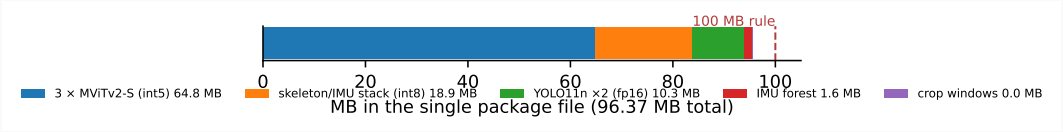
Core insight

Complementary small sensors plus recording structure beat model size.

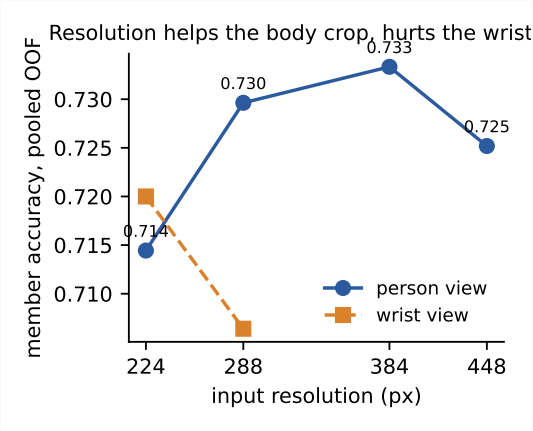
- every larger single model overfit 2 281 training clips (VideoMAE-B fine-tune: 0/4 folds)
- five small members, fused in the right probability space, decoded with an activity-script prior
- the gain is not only higher accuracy: it is a **flatter** per-subject distribution (sd 6.9 $\rightarrow$ 5.4)

System overview

raw clip $\rightarrow$ YOLO11n person window / YOLO11n-pose wrists (packaged) $\rightarrow$ IR+Depth crops 384 / 224 px, thermal full frame 224 px $\rightarrow$ 3 $\times$ MViTv2-S (int5) $\parallel$ skeleton/IMU stack (20 members, int8) $\parallel$ IMU-statistics forest $\rightarrow \sum w_i \log p_i$ (prior-corrected) $\rightarrow$ ordered transition decoder $\rightarrow$ CSV



Seeing the person and the hand: the crop-size rule



- person crop: 224 → 288 → 384 px, +1.5 and +0.4 points, 4/4 folds; 448 px: −17 fused clips
- wrist crop: 224 → 288 px is −1.36 points
- resolution pays where the crop is large relative to the source and costs where the crop is already upsampled

Thermal comes back

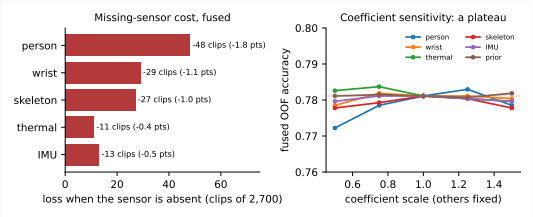
- at 128 px, cropped: member 0.544, **zero** fusion gain at any coefficient
- the same recipe at 224 px, full frame: member 0.627 (+5.6, 4/4 folds), +46 of 2 700 fused clips, +5 public
- the failure was the recipe, not the sensor: heat of objects and scene context complement the infrared crop

Skeleton and inertial members

family	members	bytes
adaptive-graph ST-GCN, ST-GCN, joint/velocity/bone TCN	12	18.9 MB int8
invariant-feature IMU TCN, world-frame IMU TCN	8	
IMU statistics forest (545 features, 200 trees)	1	1.6 MB

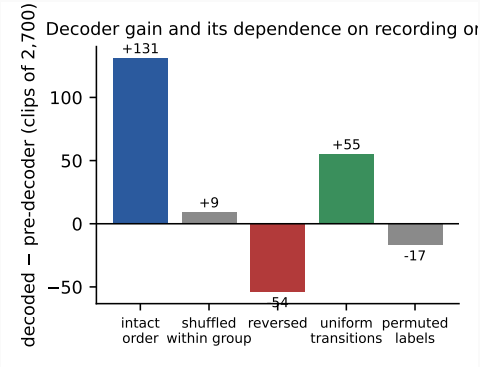
- removing the skeleton stack costs 27 clips of 2 700; the forest 11 (neutral, kept on a paired test)
- no absolute orientation from the IMUs: it is a subject/mounting leak

Fusing in the right space



- video members emit uniform-prior posteriors, the skeleton stack train-prior posteriors: divide before fusing (wrong direction: -4 points)
- coefficients by nested cross-validation; every coefficient $\pm 50\,\%$ stays within $\pm 1\,\%$
- any single absent sensor costs ≤ 1.8 points

Structure as a prior: the ordered transition decoder



- recordings are scripted; clips are grouped by their radar session key and ordered in time
- transition model fitted on training metadata only; beam search; soft distinctness penalty
- $+131$ clips intact; $+9$ if the order is shuffled; $+55$ from distinctness alone; harmful only if reversed

What generalises

- 18-subject leave-one-subject-out: person view mean 0.726, sd 0.069, spread 28 points
- shipped ensemble: mean 0.782, sd **0.054**, spread **21 points**; standard error of an 8-subject draw 2.44 $\rightarrow$ 1.92
- a lower-quartile rule that re-ranks subjects per candidate rewards shuffling; we use a fixed subject set and paired deltas

Fitting 100 MB

- three MViTv2-S as 5-bit weight codes, bit-packed: 25.9 MB each (int5 vs int6: one public clip, recovered by the thermal soup)
- skeleton stack int8 + deflate 22.8 $\rightarrow$ 18.9 MB; forest as tensors 1.6 MB
- detectors fp16, 10.3 MB: int8 moved 150 of 405 crop windows, so they are not quantised
- 96.37 MB on disk; nothing outside the file is loaded at inference

What did not work

frozen foundation teacher (V-JEPA 2)	0.419 vs 0.712
bigger backbones	overfit, 0/4 folds
per-subject alignment, object channel, pair verifier	dead (purity 0.34, AUC 0.50, AUC 0.59)
seed soups, temporal TTA, full-frame IR+Depth	-2.4, fusion loss, correlated copy
transductive self-training	member $+1.4$, fusion -2
person view at 448 px	-17 of 2 700

One cause: the residual error is per-subject and correlated; each lever assumes independent evidence.

Reproducibility

- `bash inference.sh <data_dir> <out.csv>`: byte-identical to the submitted file on an RTX 4060 laptop and a cluster A100, and in a network-isolated container
- five GPU generations (RTX 4060, A100, H100, H200, V100): max posterior difference $\leq 2.0 \times 10^{-3}$, prediction changes 0/405 on each, CSV identical — unchanged predictions is the claim, not bit-equality
- hostile synthetic set: reused ids, new ids, missing sensors, malformed files, shuffled or absent submission template — completes, rows in order
- disclosed residual: the primary CSV was fused before the skeleton stack was pruned to fit 100 MB — 403/405 rows, two decoder near-ties, documented

Results

	clips	accuracy
Kaggle public (4 subjects)	174/201	0.866
Kaggle private (same subjects)	179/204	**0.877** (14th of 307)
pooled out-of-fold, decoded (16 subjects)	2 240/2 700	0.830
organisers' held-out set	to be published	

Takeaway

The next gain is more subjects, not more model.

- the members hold the information (oracle over five members 0.877); selection under subject shift is the barrier
- the subject learning curve is flat past six subjects
- we froze the competitive solution, reproduced it exactly, stress-tested it on unseen data, and showed which parts mattered